WORKED SOLUTION

STEP II 2014, Question 3

Pure • Coordinate geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s2-q3

The key idea

The tangent at P is just a straight line, with gradient y' and y-intercept $y - xy'$, so part (i) turns the constant-distance property into a differential equation. Differentiating that equation makes a factor y'' appear, and the two ways of making the product zero give the two families of curves: straight lines ($y'' = 0$) and a circle. Part (iii) then asks us to glue pieces of these together smoothly.

Solution

Part (i)

If $m = 0$ the line is $y = c$ and the distance is clearly c, so suppose $m \neq 0$. The line meets the y-axis at $A = (0, c)$, and the line makes an acute angle θ with the x-axis, where $\tan\theta = |m|$. Let N be the foot of the perpendicular from O to the line. In the right-angled triangle ONA, the angle at A between the line and the y-axis is $\frac{\pi}{2} - \theta$, so the angle AON is θ, and

$$ON = OA\cos\theta = \frac{c}{\sqrt{1+\tan^2\theta}} = \frac{c}{\sqrt{1+m^2}} = c(m^2+1)^{-\frac{1}{2}}.$$

(The same value comes from minimising $x^2 + (mx+c)^2$, a quadratic in x with minimum value $\frac{c^2}{1+m^2}$ at $x = -\frac{mc}{1+m^2}$.)

If $c < 0$, reflecting in the origin turns the line into $y = mx + |c|$ without changing distances from O, so in general the distance is $|c|(1+m^2)^{-\frac{1}{2}}$.

Part (ii)

At $P = (x, y)$ the tangent has gradient y' and passes through P, so its equation in coordinates (X, Y) is

$$Y = y'X + (y - xy').$$

This is a line of the form in part (i) with $m = y'$ and $c = y - xy'$ (it is non-vertical by assumption). Its distance from O is therefore

$$a = \frac{|y - xy'|}{\sqrt{1 + (y')^2}},$$

and squaring gives

$$(y - xy')^2 = a^2(1 + (y')^2). \qquad (*)$$

Differentiating. Here a is a constant, and $\frac{\mathrm{d}}{\mathrm{d}x}(y - xy') = y' - y' - xy'' = -xy''$. So differentiating $(*)$ gives

$$2(y - xy')(-xy'') = 2a^2y'y'', \qquad \text{that is,} \qquad y''[\,x(y - xy') + a^2y'\,] = 0.$$

Ask yourself How does a product being zero turn into two separate cases for the curve?

At each point, either $y'' = 0$ or the bracket is zero. On a stretch of the curve where $y'' = 0$ the curve is straight; on a stretch where $y'' \neq 0$ the bracket must vanish. We take the two cases in turn. (We assume $a > 0$; if $a = 0$ every tangent passes through O and C is simply a line through the origin.)

Case 1: $y'' = 0$. Then $y' = m$ is constant and $y = mx + k$ for constants m, k. Substituting into $(*)$, where $y - xy' = k$, gives $k^2 = a^2(1 + m^2)$, so

$$y = mx \pm a(1 + m^2)^{\frac{1}{2}}.$$

Case 2: the bracket is zero. Then $xy - x^2y' + a^2y' = 0$, so

$$(x^2 - a^2)\,y' = xy.$$

First, $x^2 \neq a^2$: if $x = \pm a$ then $xy = 0$ forces $y = 0$, and $(*)$ becomes $a^2(y')^2 = a^2(1 + (y')^2)$, which is impossible. So we may divide:

$$y' = \frac{xy}{x^2 - a^2}, \qquad y - xy' = \frac{y(x^2 - a^2) - x^2y}{x^2 - a^2} = \frac{-a^2y}{x^2 - a^2}.$$

Substituting both into $(*)$ and multiplying through by $(x^2 - a^2)^2/a^2$:

$$\begin{aligned} & a^2y^2 = (x^2 - a^2)^2 + x^2y^2 \\ \iff \quad & y^2(a^2 - x^2) = (a^2 - x^2)^2 \\ \iff \quad & (a^2 - x^2)\,(x^2 + y^2 - a^2) = 0. \end{aligned}$$

Since $a^2 - x^2 \neq 0$, we conclude $x^2 + y^2 = a^2$.

So either $y = mx \pm a(1 + m^2)^{\frac{1}{2}}$ for some m, or $x^2 + y^2 = a^2$, as required. Both do have the property: the lines are at distance a from O by part (i), and every tangent to a circle centred at O with radius a is at distance a from O.

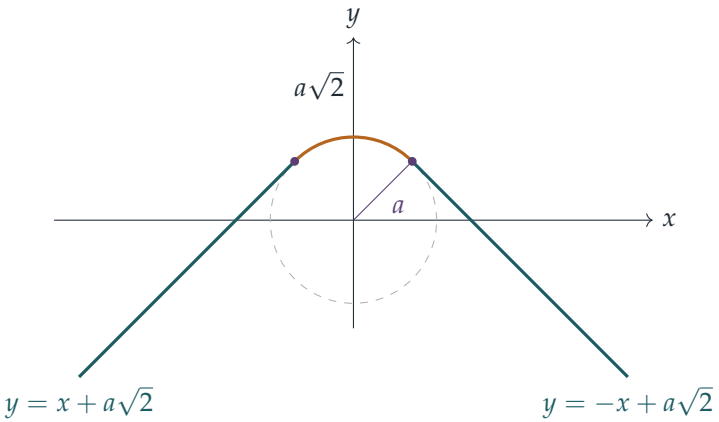
Part (iii)

At a point where a straight piece meets the arc, C must have a tangent, so the line must have the same gradient as the arc there: each line is the tangent to the circle at the end of the arc. Also, every tangent must be non-vertical, so the arc must not reach the points $(\pm a, 0)$, and since C is the graph of a function of x it uses only the upper or only the lower half of the circle.

An example: take the arc of $x^2 + y^2 = a^2$ from $(-\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}})$ to $(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}})$ over the top, continued by the tangents at its ends:

$$y = \begin{cases} x + a\sqrt{2}, & x \le -\frac{a}{\sqrt{2}}, \\ \sqrt{a^2 - x^2}, & |x| \le \frac{a}{\sqrt{2}}, \\ -x + a\sqrt{2}, & x \ge \frac{a}{\sqrt{2}}. \end{cases}$$

The lines have $m = \pm 1$ and intercept $a\sqrt{2} = a(1+m^2)^{\frac{1}{2}}$, as Case 1 requires. At $x = \pm\frac{a}{\sqrt{2}}$ the arc has gradient $-\frac{x}{y} = \mp 1$, matching the lines, so the gradient is continuous and finite everywhere.



Any arc of the upper (or lower) semicircle not containing $(\pm a, 0)$, continued by its end tangents, works in the same way.

Checks

- In part (i), $m = 0$ gives distance c, and as $|m| \to \infty$ the line becomes steep and the distance tends to 0, as it should for a line through $(0, c)$.
- On the circle, $y' = -x/y$ gives $y - xy' = a^2/y$, and both sides of $(*)$ equal a^4/y^2.
- For the example with $a = 1$, the distance from O to the tangent was checked by computer at 1001 points between $x = -5$ and $x = 5$ and was 1 every time; at $x = 2$, for instance, the tangent is $y = -x + \sqrt{2}$, at distance $\sqrt{2}/\sqrt{2} = 1$.

Common slips

- Dividing the differentiated equation by y'' and losing the straight-line solutions.
- Forgetting that a is constant, or differentiating xy' as y' instead of $y' + xy''$.
- In part (iii), joining the lines to the arc at a corner, or using a whole semicircle, whose ends have vertical tangents.

Beyond STEP

The circle here is the *envelope* of the family of lines $y = mx \pm a\sqrt{1+m^2}$, and $(*)$ is a Clairaut equation, $y = xy' + f(y')$: its general solution is a family of straight lines and its *singular solution* is their envelope. As a quadratic in y', $(*)$ has discriminant $4a^2(x^2 + y^2 - a^2)$, so the two gradient choices merge exactly on the circle; that is why solutions can be patched together there, as in part (iii), and why uniqueness of solutions fails. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness for first-order equations.

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